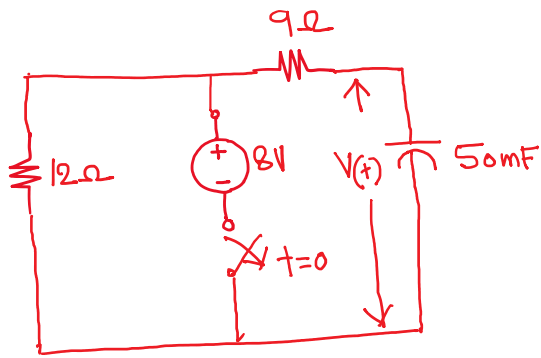
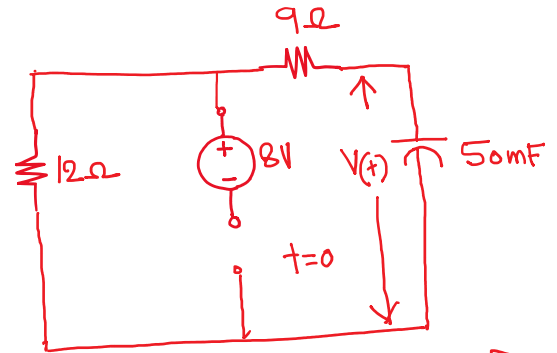
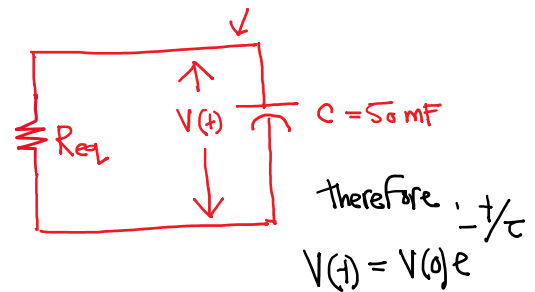




steps: for $t \geq 0$, the ckt looks like:



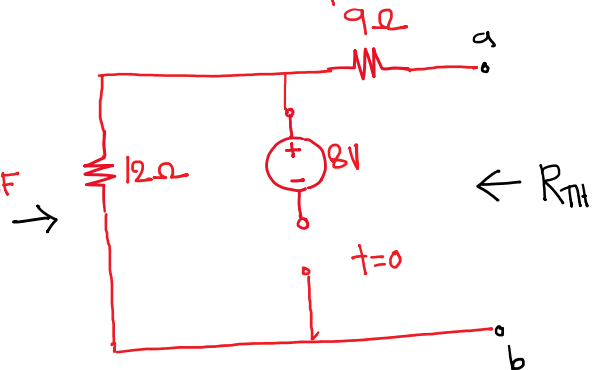
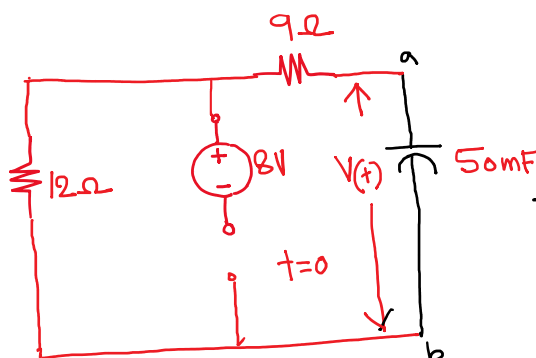
Represent the ckt in the following form for $t \geq 0$



$$\tau = R_{eq}C$$

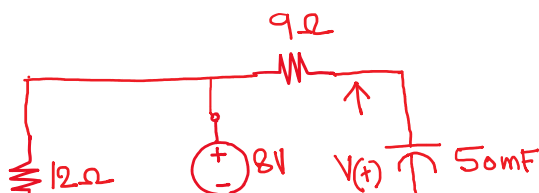
$R_{eq} = R_{TH}$ for the ckt $t > 0$, which is obtained by using the following steps:

1. Remove the capacitor.
2. Find R_{TH} looking through open terminals after the capacitor has been removed.

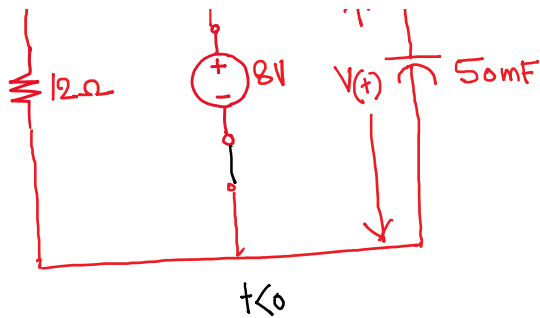


$$R_{eq} = R_{TH} = 12 + 9 = 21 \Omega$$

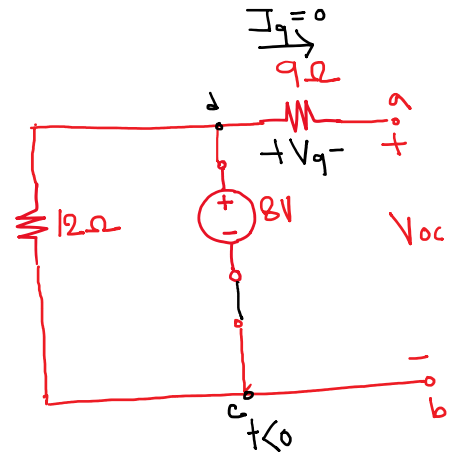
For determining $V(0)$, find $V(0^-)$ for $t < 0$.



For $t < 0$, Capacitor is open for the DC voltage source



$V(0^-)$



Apply KVL at $bcdab$ loop:

$$+8 - V_g - V_{OC} = 0$$

$$8 - 9I_g - V_{OC} = 0 \text{ [ohm's law]}$$

$$V_{OC} = 8V \text{ [} I_g = 0, \text{ open terminal]}$$

$$V(0^-) = V_{OC} = 8V$$

Voltage across capacitor doesn't change abruptly:

$$V(0^+) = V(0) = V(0^-) = V_{OC} = 8V$$

$$V(0) = V(0^+) = V_{OC} = 8V$$

therefore, t/τ

$$\begin{aligned} V(t) &= V(0)e^{-t/\tau} \\ &= 8e^{-\frac{t}{1050 \times 10^{-3}}} \text{ V.} \\ &= 8e^{-0.952t} \end{aligned}$$

$$\tau = R_{eq}C = 21 \times 50 \times 10^{-3} \text{ sec}$$

$$V(0) = 8V$$

